Frozen VLM inference

Language model

Tokenizer

Question Q

What surname is printed above the number 4?

X_S

Q H **ReadPrune**

X

Projector

Vision encoder



Frozen modules

ReadPrune question-conditioned, training-free token selection

1 Readout influence

Final-block input

H

Cached

Intact

Ablate i; renorm.

Shared

Φ_H

CLS tail

e

e_{-i}

Alignment change

$\delta_i = t^{\mathrm{T}}(e - e_{-i})$

t

Q

CLIP text encoder

Readout influence $|\delta|$

2 Influence-guided selection

Native-feature diversity

X → Regularized cosine → B

Selection weights

$|\delta|$ → Bounded mapping → q

Kept: 64 / 576

Greedy DPP $(|S| = K)$

$L = \mathrm{diag}(q^2)\, B\, \mathrm{diag}(q^2)$

S

Raster-order gather X_S

